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(SECP3133) HIGH PERFORMANCE DATA PROCESSING

SECTION 02

REAL WORLD CASE STUDY EXERCISE: SHOPEE

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Part A: Case Study Analysis – Shopee

A1. Data Characteristics

Shopee deals with vast amounts of data daily for millions of users, sellers and transactions. The key services provided by the platform are mainly batch data, streaming data and real-time data. Real-time processing is particularly crucial for the following features: product recommendations, payment updates, order tracking and interactions with Shopee Live. Most of the data sources are mobile apps, websites, seller systems, payment gateways, logistics providers, customer reviews and user activity logs. The 3Vs of big data can be used to explain the data characteristics of Shopee:

- Volume - a lot of transaction data, product data, images, videos, customer data
- Velocity – data created and handled very rapidly based on user click, search, purchase and live activity
- Structured data – transactions, user profiles
Unstructured data – images, videos, reviews, chats
Both data are all handled on the platform

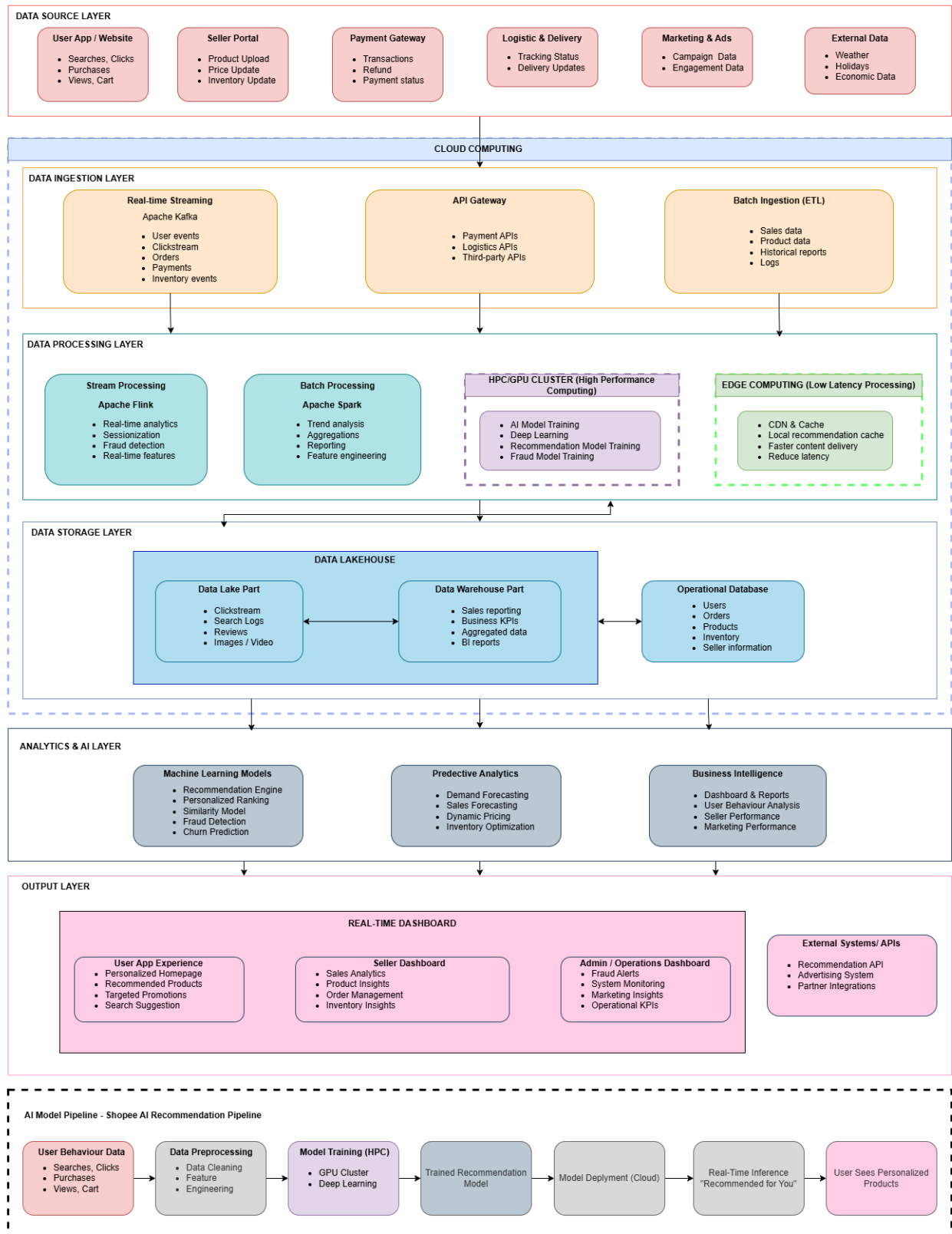
A2. Key Challenges

One major challenge faced by Shopee is scalability. When users use the system in campaigns like 11.11 or 12.12, the number of users and transactions skyrockets requiring rapid scaling of resources. One of the other challenges is low latency processing. The recommendation system, payment verification, order tracking and other features need to be responsive in real time to deliver a positive user experience. High-speed data processing and data management in storing are also challenges for Shopee. It is receiving massive amounts of data that need to be efficiently processed and stored to enable data analytics, AI models and business operations. Data security and privacy are also crucial when dealing with the payment system and sensitive customer information, as Shopee handles such data.

A3. Technologies Required

Shopee has implemented AI and Machine Learning (ML) technologies to personalize recommendations, target ads, prevent fraud and analyze customer behaviour. If the data streams are fast and real-time streaming is required, then technologies like Apache Kafka would be appropriate for collecting and distributing the data streams in real-time, in a very efficient manner. Large-scale analytics and real-time data processing are made easier by distributed processing systems such as Apache Spark and Apache Flink. Shopee can leverage cloud-based data lakes or data lakehouses to efficiently store and manage structured and unstructured data, as well as power analytics and AI workloads for storage.

Part B: Architecture Design



Part C: Justification

C1: Architecture Justification

1. Data Source Layer

The data source layer is designed to collect data from multiple channels within the Shopee ecosystem. This includes user activity from the mobile application and websites, seller portal updates, payment gateway information, logistics and delivery tracking, marketing campaigns and external contextual data.

This layer is important because Shopee relies heavily on customer behaviour data to understand user preferences and improve recommendation accuracy. User actions such as searches, clicks, purchases, cart activity and browsing patterns provide valuable insights into customer interests and shopping behaviour. At the same time, seller and logistics data ensure product availability and accurate order tracking. By integrating multiple data sources, Shopee can gain a more complete understanding of both customer behaviour and operational performance.

2. Data Ingestion Layer

The data ingestion layer ensures Shopee can collect and transfer large volumes of data efficiently. Apache Kafka is selected for real-time streaming because Shopee continuously generates high-speed events such as searched, clicks, purchases and inventory updates. Batch ingestion (ETL) is used to process historical sales data, reports and logs for long-term analysis. Moreover, an API gateway supports smooth communication between Shopee and external services such as payment and logistics providers.

3. Processing Layer Justification

The data processing layer transforms raw data into meaningful insights for Shopee's recommendation system and business analytics. Apache Flink is used for stream processing to support real-time activities such as fraud detection and recommendation updates. Apache Spark processes historical data for reporting and trend analysis. HPC using GPU clusters accelerates AI model training, while edge computing improves response speed through caching and CDN services to reduce latency and enhance user experience.

4. Storage Layer

A data lakehouse architecture is chosen because Shopee manages both structured and unstructured data at large scale. The data lake component stores raw data such as clickstream logs, searches, reviews and multimedia content. Meanwhile, the warehouse component stores structured information such as sales reports and business KPIs. Operational databases are also included to handle transactional records such as users, products, orders and inventory efficiently and reliably.

5. Analytics and AI Layer

The analytics and AI layer helps Shopee improve customer experience and business performance through intelligent decision-making. Machine learning models are used for personalized product recommendations, ranking systems and fraud detection. Predictive analytics supports demand forecasting and inventory planning to improve operational efficiency. In addition, business intelligence tools help Shopee analyse customer behaviour, marketing performance and seller activities, enabling the company to make informed, data-driven decisions.

6. Output Layer

The output layer is responsible for delivering processed insights to different users through real-time dashboards and application interfaces. Customers receive personalized recommendations, promotions, and search suggestions to improve their shopping experience. Sellers can monitor product performance, sales, and inventory status, while administrators track fraud alerts, campaign performance, and operational KPIs. This allows Shopee to respond quickly to business changes and maintain platform efficiency.

7. AI Model Pipeline

The AI model pipeline is included to demonstrate how Shopee's recommendation system functions from training to deployment. User behaviour data such as searches, clicks and purchases are first cleaned and prepared through preprocessing. HPC GPU clusters are then used to train recommendation models efficiently. After training, the model is deployed in the cloud to personalized products to customers.

8. Data Lakehouse

A data lakehouse architecture is included because Shopee requires a system that can support both raw big data storage and structured analytics. The data lake component stores flexible, large-scale user behaviour data, while the warehouse component supports reporting and business intelligence. Combining both systems into one platform improves scalability, reduces data duplication and enables Shopee to process analytics more efficiently for recommendation systems and business reporting.

9. Real-Time Dashboard

Real-time dashboards are added to help Shopee visualize processed data and support immediate decision-making. During major sales campaigns and flash sales, Shopee can monitor customer traffic, sales performance, fraud detection, and operational activities instantly. Different stakeholders, including customers, sellers, and administrators, can access relevant insights through dashboards. This improves operational efficiency, enhances system monitoring, and helps Shopee maintain a smooth user experience.

C2: Computing Paradigm Justification

Proposed Shopee architecture involves a cloud computing, edge computing and high-performance computing (HPC) based hybrid computing paradigm. The approach is applicable since Shopee handles large scale real-time e-commerce traffic which demands scalability, low latency and fast analysis.

Cloud computing is adopted as a key infrastructure due to its ability to offer scalable storage, distributed processing and flexible resource management. It also facilitates the data lakehouse architecture, the deployment of AI models and real-time dashboards and enables Shopee to manage traffic surges during its campaign, such as 11.11 and 12.12 sales.

Edge computing is also connected via content delivery network (CDN) caching and local edge servers to shorten latency and make sure that the response time of services like Shopee Live, payment transaction or product loading are smooth. This enhances the user experience and system performance. For tasks like training machine learning models, recommendation systems, fraud detection, which required a lot of computation, HPC with graphic processing unit (GPU) clusters is needed. By utilizing parallel processing, Shopee can process large datasets quickly and efficiently and train AI models faster.

In summary, the hybrid computing paradigm helps Shopee to scale up its operations, deliver quick response times and process large amounts of data efficiently in real time.